

Consumer Preferences in the Used Vehicle Market: Willingness to Pay for Powertrain Type, Driving Range, Vehicle Condition, and Operating Cost

Zain Hoda¹, Xiatian Iogansen^{1,2}, Christina Gore², Joshua D. Kneifel², Sindhu Ranganath^{1,2}, John P. Helveston¹

¹Department of Engineering Management and Systems Engineering, George Washington University

²Applied Economics Office, Engineering Laboratory, National Institute of Standards and Technology

Background

Used vehicles account for roughly 70% of U.S. vehicle transactions (Bureau of Transportation Statistics, 2023), yet most alternative fuel vehicle (AFV) research focuses on new-vehicle buyers. This study estimates willingness to pay (WTP) for powertrain type and other attributes among prospective U.S. used-vehicle buyers and compares WTP with real-market used-vehicle prices.

Data and Method

An online survey (Dynata and Prolific, December 2025–April 2026) of U.S. adults planning to buy a used car or SUV within two years yielded 2,254 respondents. Each completed six discrete choice questions varying powertrain (conventional, gas hybrid [HEV], battery electric [BEV]), price, operating cost, BEV range, model year, and mileage. Six mixed logit models, stratified by car/SUV segment and then by budget tier (low/high), were estimated in WTP space. Simulated net WTP was also compared with real-market price premiums for matched used vehicle pairs (AFV vs conventional).

Results

Mean BEV utility is negative and highly significant in all models (-\$10.4k to -\$26.1k), with large standard deviations indicating within-segment heterogeneity. HEVs carry a significant premium only among high-budget buyers (about \$4,200 for cars, \$2,300 for SUVs). The BEV penalty narrows sharply with range, from about -\$8,200 to -\$18,800 at 100 miles to -\$2,600 to -\$11,700 at 300 miles (Figure 1); only high-budget car buyers considering a 300-mile BEV are statistically indistinguishable from the conventional baseline. Mileage, age, and operating cost matter less than powertrain and range. Observed used-BEV premiums exceed the WTP-implied parity requirement in every matched pair (by about \$6,100 for the high-budget car, \$15,700 for the low-budget car, and \$16,900 for the low-budget SUV), while HEV results are mixed. The former \$4,000 used clean vehicle credit could cover the gap for high-budget car buyers considering longer-range BEVs, but its \$25,000 price cap excludes most longer-range used BEVs. These results favor segment-aware policy, including non-price instruments such as battery health disclosure.

Willingness to Pay for Vehicle Attributes

By vehicle segment and budget tier; error bars show 95% confidence intervals

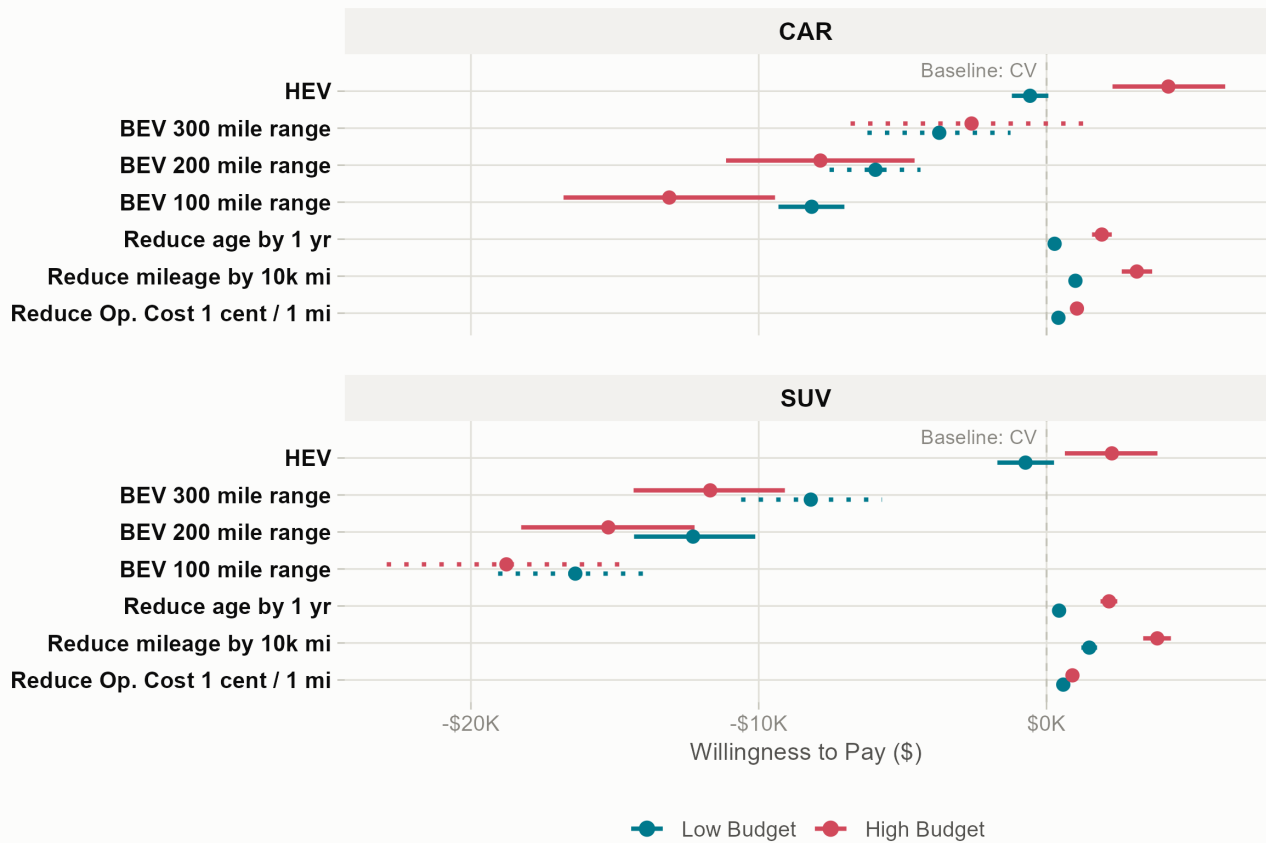


Figure 1: Willingness to pay for vehicle attributes, by vehicle segment and budget tier (95% confidence intervals shown; values in thousands of dollars). For the BEV range configurations, solid intervals mark WTP calculated within the range shown to that segment and dotted intervals mark WTP extrapolated beyond it.

Consumer Valuation of Battery Health Attributes in Used Battery Electric Vehicle Markets: A Discrete Choice Experiment

Xiattian Iogansen^{1,2}, Zain Hoda¹, Christina Gore², Joshua D. Kneifel², Sindhu Ranganath^{1,2}, John P. Helveston¹

¹Department of Engineering Management and Systems Engineering, George Washington University

²Applied Economics Office, Engineering Laboratory, National Institute of Standards and Technology

Background

Used vehicle transactions account for roughly 70% of U.S. passenger vehicle purchases (Bureau of Transportation Statistics, 2023), and used BEVs are entering the secondary market at an accelerating pace (Cox Automotive Inc., 2025). Despite this, 52% of U.S. adults are unlikely to consider a used BEV purchase (Fernandes, 2023), primarily citing uncertainty about battery condition. This information gap suppresses demand and prevents efficient pricing in the secondary market.

Data and Method

We administered an online discrete choice experiment (December 2025–April 2026) to 3,072 U.S. adults intending to purchase a used car or SUV within two years. Each respondent completed six choice tasks among three hypothetical used BEVs or a no-choice option, varying mileage, price, electric range at Year 3, annual battery degradation rate, and battery refurbishment history. The design was stratified by body type and budget, and half the sample received extended battery cost and warranty information. We estimate a six-class Latent Class Choice Model on 2,916 respondents with complete covariate data (17,496 choice observations); class-specific WTP is the negative ratio of the attribute coefficient to the price coefficient.

Results

The six classes differ markedly in battery quality sensitivity, price responsiveness, and market engagement; the attached figure (Figure 2) shows more details on each class.

Class 1: Battery Health Attentives (22.7%). Highest range loss aversion (about \$2,300 per percentage point) and clear refurbishment penalties.

Class 2: Multi-Attribute Maximizers (20.8%). Highest premium for a minimum viable range (\$11,830) and large refurbishment penalties (about -\$6,000).

Class 3: Range-Focused, EV-Knowledgeable (16.6%). Very high WTP for range below 130 miles (\$38,700 per 100 miles); negligible refurbishment penalties.

Class 4: Budget-Constrained, Low-WTP (12.8%). Largest price sensitivity and modest WTP for any attribute, yet 91.2% never opt out.

Class 5: Opt-Out Dominant, BEV-Skeptical (8.7%). 75.4% chose no vehicle in all six tasks; most WTPs are indistinguishable from zero.

Class 6: Low-Engagement (18.5%). Near-zero price coefficient, consistent with attribute non-attendance; WTP estimates treated as unreliable.

Classes 1 and 2 (43.5%) prioritize battery health and are the primary market for standardized battery health certification.

Bureau of Transportation Statistics (2023). *National Transportation Statistics*. Tech. rep. U.S. Department of Transportation, Bureau of Transportation Statistics.

Cox Automotive Inc. (2025). *EV Market Monitor – February 2025*. (Visited on 06/30/2025).

Fernandes, J. (2023). *US: Understanding Consumer Opinion about Used Electric Cars*. (Visited on 06/30/2025).



Figure 2: WTP and class profile summary.